

Is this RAG system good?

An engineering assessment · judgement, not a new measurement · every figure read from the project's own artifacts

VERDICT

The engineering discipline is excellent — better than most production RAG work. The retrieval numbers are decent. But the cross-encoder is probably not paying for itself, and that matters because it is 91.5% of your runtime.

The discipline is excellent and the V1 holdout number is real. The risk is not sloppiness - it is that the rigour is being spent on differences too small to be real, and that the most expensive component has the weakest evidence behind it.

87.8%

V1 holdout strict R@10, 79/90, unseen

11.5×

latency the CE costs, for +1 case of 20

-0.0070

MRR the CE costs versus not using it

202

documents · 14,209 chunks

What is genuinely good

The V1 holdout number is real. `SYSTEM-D-GUARD-BLEND` scored **79/90 = 87.8%** strict Recall@10, span recall 0.8833, document recall 0.9778, MRR 0.7055 — on a split frozen contamination-aware rather than randomly, guarded by an access log, and opened once. Most teams never get a number that clean.

The process is the strongest part of this project. Each of these was checked against the artifacts rather than taken on trust:

- preregistration written before any scores exist
- frozen system config hashes; a changed field changes the hash
- corpus snapshot reproduces to an exact manifest hash
- contamination-aware split, not random; holdout access-logged
- EVAL-VAL-001 rejected SYSTEM-B when it failed to replicate
- EXP-015 refused to substitute a hand-rolled reranker rather than fake a result
- EXP-018B Track 1 proved score-preservation bitwise, not by tolerance

Two deserve singling out. **EVAL-VAL-001 rejected SYSTEM-B when it failed to replicate** — killing your own promising result is rare. And **EXP-015 refused to substitute a hand-rolled scorer** when no pretrained cross-encoder was reachable, rather than produce a number that would have looked fine.

The thing I would worry about

“Reranker rejected at development” and “reranker is 91.5% of runtime” cannot both be fine, so I read EXP-016. Development, n=20:

variant	strict R@10	MRR	latency ms	net rescues
A — no cross-encoder	19/20	0.8309	500.8	—
B — CE rerank (EXP-015 control)	18/20	0.8030	5,780.2	-1
C — guard	20/20	0.8072	5,774.4	+1
D — guard + 0.7/0.3 blend (shipped) shipped	20/20	0.8239	5,774.4	+1

The cross-encoder buys **+1 case out of 20**, at **11.5× the latency**, and it **lowers MRR** against not using it at all (0.8309 → 0.8239).

Now read what the guard is actually doing. These are cases SYSTEM-A already had right, which the cross-encoder pushed out of the top 10:

case	A rank	CE rank	D rank	clamped by guard
HA-24	1	18	3	yes
HA-22	2	21	6	yes

THE GUARD'S JOB IS TO PROTECT A'S RANKING FROM THE RERANKER

EXP-015 recorded the honest verdict first: `RERANKER_REJECTED-at-dev`, net -1 — one rescue (GOLD-B005-11) against two regressions (HA-22, HA-24). The component then returned as a guarded blend. The net is one case gained and MRR lost, for 5.3 seconds.

That makes PERF-002 slightly beside the point. Making a 5,904 ms component 1.9× faster matters less than asking whether you need it.

C1 — HIGH — THE MISSING ABLATION

I could not find **D versus D-without-CE on V2-DEVSET-001 (n=50)**. EXP-016's comparison is n=20. If the cross-encoder's contribution on the larger devset is also one or two cases, the correct move is to **cut it, not optimise it**: you would get roughly 550 ms of retrieval instead of 6,455 ms, and lose about one case.

This is one run, and it is the highest-information thing available right now.

Two structural caveats

C2 — HIGH — N IS TOO SMALL FOR THE DECISIONS

EXP-018 is **+2 cases on n=50**. EXP-016 is **+1 on n=20**. Your own `systems.py` says it plainly: “*six of the last seven experiments turned on one or two cases.*”

Architecture is being chosen on deltas a bootstrap CI will not separate from noise. The 90-case holdout is the only measurement here with real statistical weight, and it is spent.

C3 — MEDIUM — THE BENCHMARK IS SOURCE-ANCHORED

Cases were mined from the corpus, then reviewed. That systematically favours extractive lexical retrieval and inflates absolute recall relative to real user questions. It is also, I suspect, why the recurring V1 finding was *“the answer is visible to the encoder but does not rank”* — that is the signature of questions written from the documents being searched.

It does not invalidate the comparisons: A versus D is fair on the same set. It does mean the headline percentage will not survive contact with real traffic.

C4 — latency is not shippable. 6,455 ms of retrieval before generation has even started; about 3,360 ms after PERF-002's bucketing; about 550 ms without the cross-encoder. **C5 — the corpus is small** (202 documents, 14,209 chunks), so the BM25 IDF and routing conclusions may not transfer at 100× the scale.

What I would do next

1. run the CE ablation on V2-DEVSET-001 before spending more on CE latency
2. get real unmined questions, even 30, from actual usage
3. treat ~6.4 s retrieval-before-generation as not shippable

TO BE FAIR TO IT

This is a research pipeline behaving like a research pipeline, and a rigorous one. The risk is not sloppiness. It is that the rigour is being spent on differences too small to be real, and that the most expensive component in the system has the weakest evidence behind it.

ASSESS-001 · an assessment, not an experiment · 0 new measurements were run for it · every figure read from `experiments/ASSESS-001/ASSESS-001-assessment.json` at build time, itself populated from EXP-015, EXP-016 and EXP-018B artifacts · the V1 holdout figures are as reported in the handoff; EVAL-HOLDOUT-001's results were not opened